BIOC0001

Lecture 2

Nucleotides and Nucleic Acids

Dr. Suzanne Ruddy

DNA Packaging- Proteins & Scaffolds

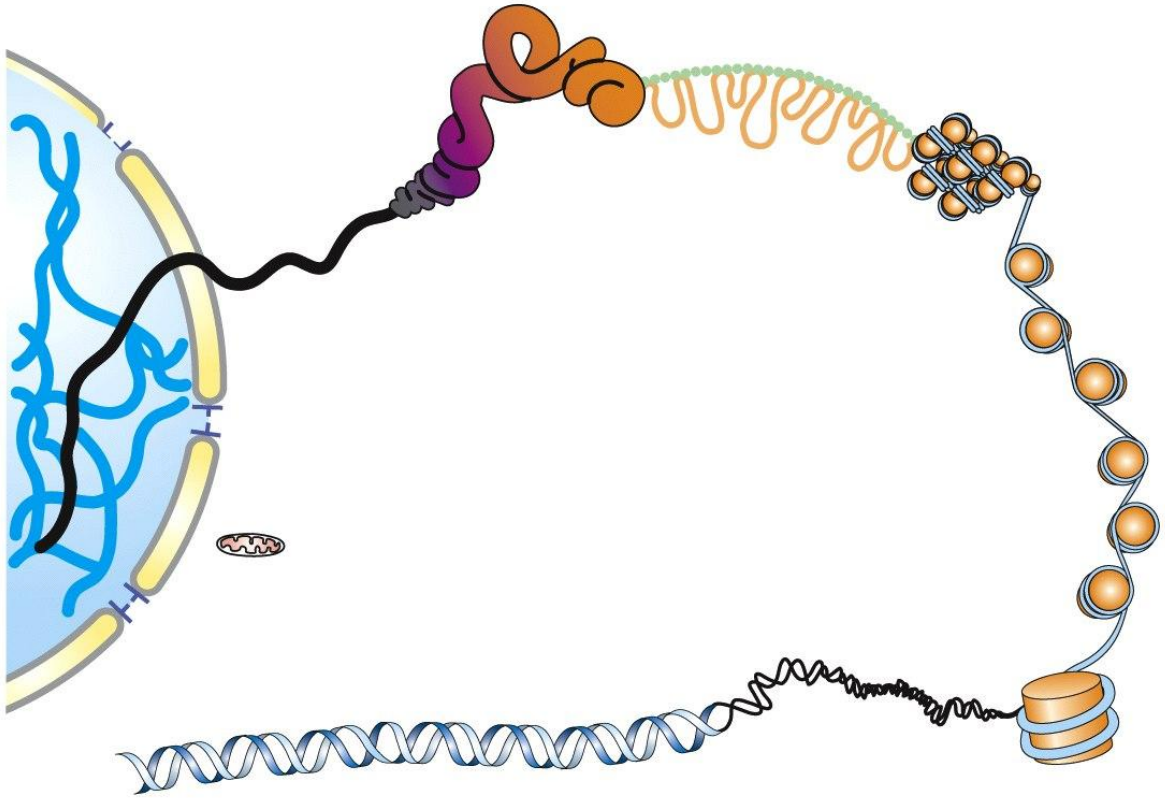


Figure 6-1
Molecular Cell Biology, Sixth Edition
© 2008 W. H. Freeman and Company

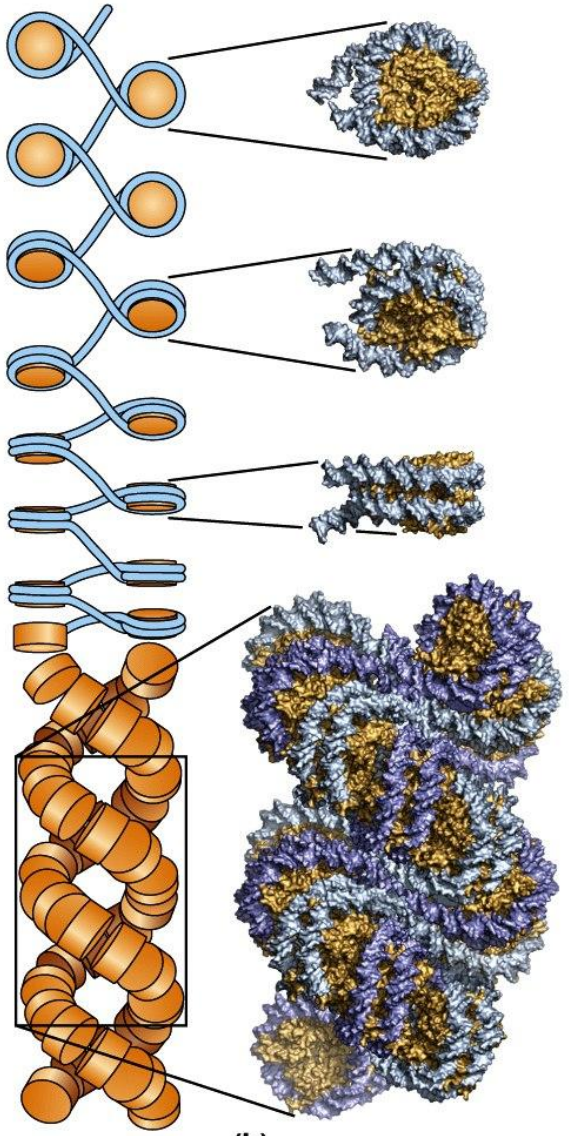
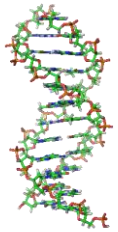


Figure 6-30
Molecular Cell Biology, Sixth Edition
© 2008 W. H. Freeman and Company

Nucleotides and Nucleic Acids: Central Dogma



DNA



RNA



PROTEIN

Deoxyribonucleic acid

- Storage form of genetic information that is transferred from one generation to the next
- Double-stranded helix
- Heteropolymer of 4 different nucleotides (G A T C)

Ribonucleic acid

- "Working copy" of the gene
- Single-stranded molecule with secondary structure
- Heteropolymer of 4 different nucleotides (G A U C)

Amino acids

- Executors of the encoded data

Polymers and Monomers

- Polymer: a molecular structure formed from a large number of similar units (monomers) linked together.
  πολυ (poly)=many; μέρος (meros)=parts
- Polymerisation: the chemical process by which two or more monomers are linked together to form a polymer
- ---

  - Polymer: nucleic acid, specifically DNA or RNA
  - Monomer: nucleotide, specifically G A T U and C

Nucleic Acids:

Ribonucleic acid (RNA).
Deoxyribonucleic acid (DNA).

Both are composed of nucleotides:
Cyclic nitrogenous bases
Pentose sugars
Phosphate groups.

The building blocks of nucleic acids: bases

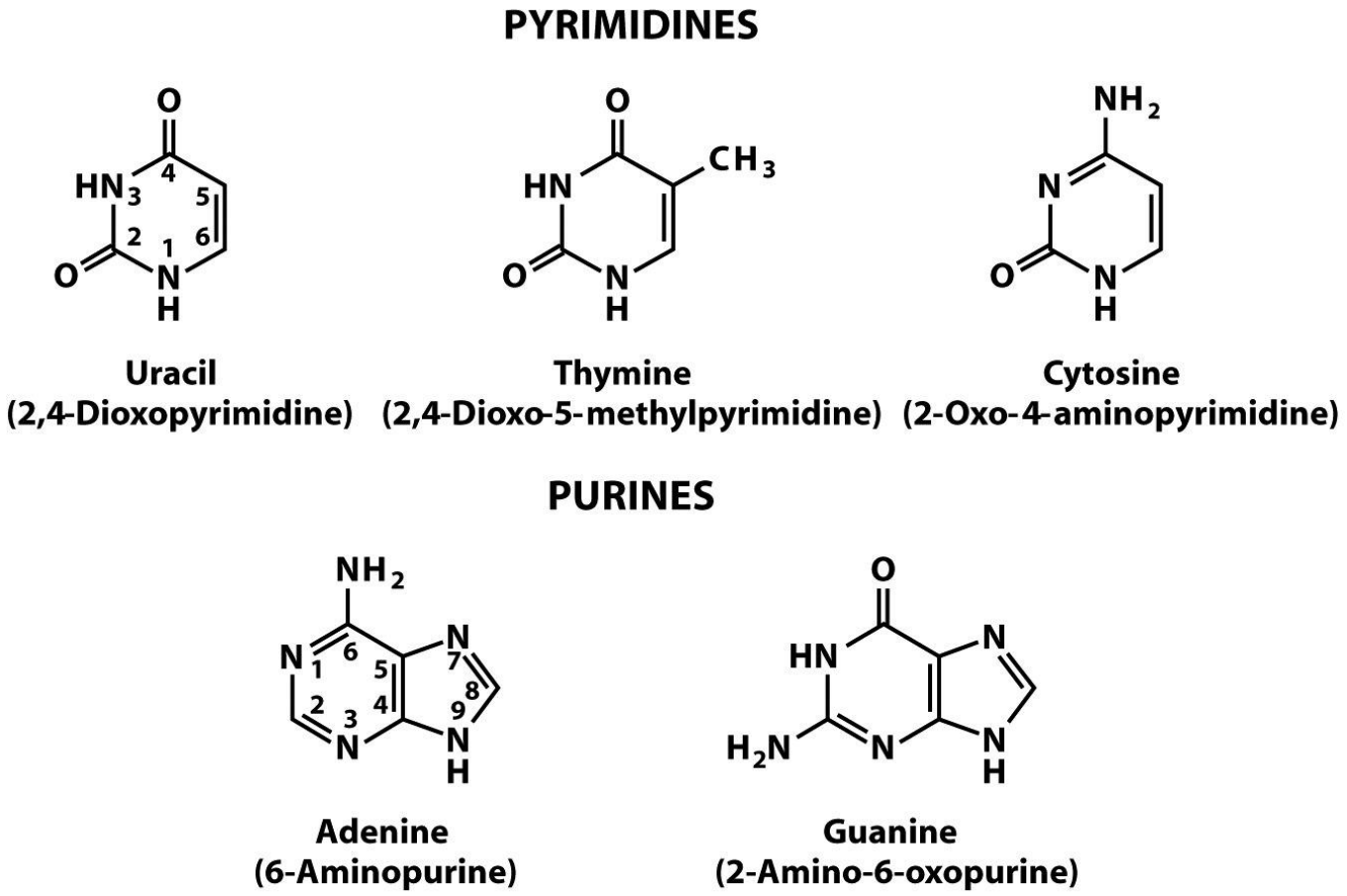
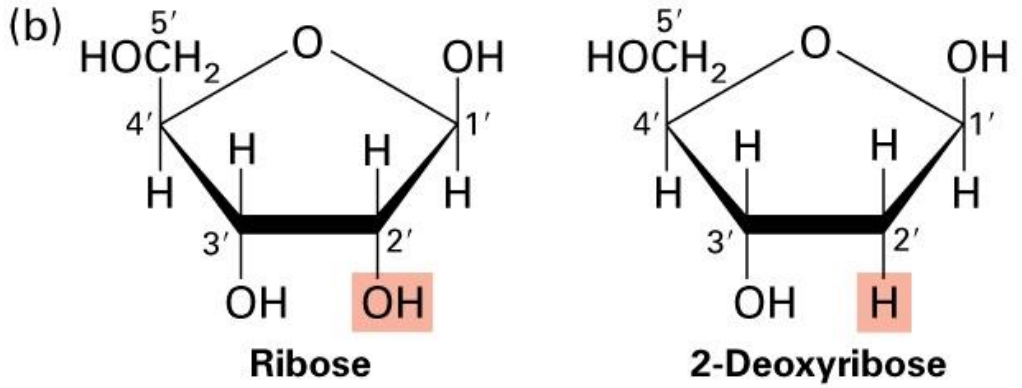


Figure 19-4 Principles of Biochemistry, 4/e
© 2006 Pearson Prentice Hall, Inc.

GUArdian Angels are PURe and TWO wINGED
(Guanine and Adenine are Purines and Two Ringed)

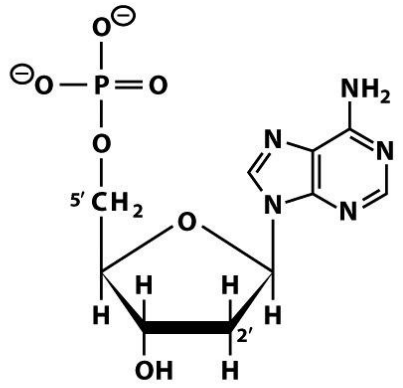
The building blocks of nucleic acids: pentose sugars



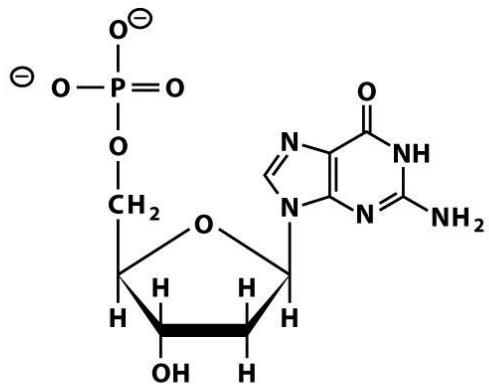
Carbons on the ribose ring are described with a prime (‘) to distinguish them from positions on the base ring.

5 membered monosaccharide ring is called a furanose ring

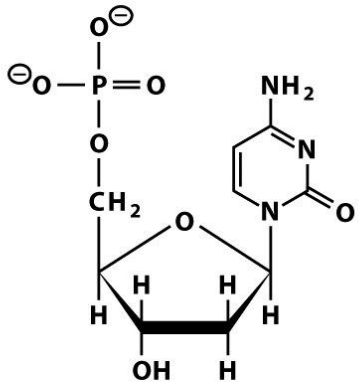
Nucleotides



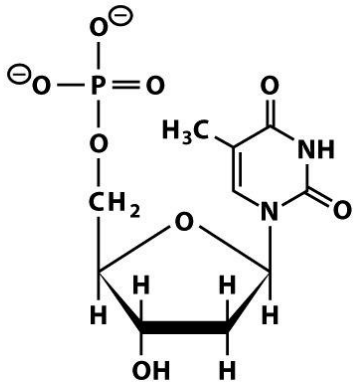
2′-Deoxyadenosine 5′-monophosphate
(Deoxyadenylate, dAMP)



2′-Deoxyguanosine 5′-monophosphate
(Deoxyguanylate, dGMP)



2′-Deoxycytidine 5′-monophosphate
(Deoxycytidylate, dCMP)



2′-Deoxythymidine 5′-monophosphate
(Thymidylate, dTMP)

Figure 19-9 Principles of Biochemistry, 4/e
© 2006 Pearson Prentice Hall, Inc.

Polarity of polynucleotide chains

- Successive (deoxy)ribose residues of a polynucleotide chain are joined by a phosphate group between the 3’ position of one sugar and the 5’ position of the next sugar.
- One end of the chain (conventionally the left) has a free 5’ end and the other end has a free 3’ end.

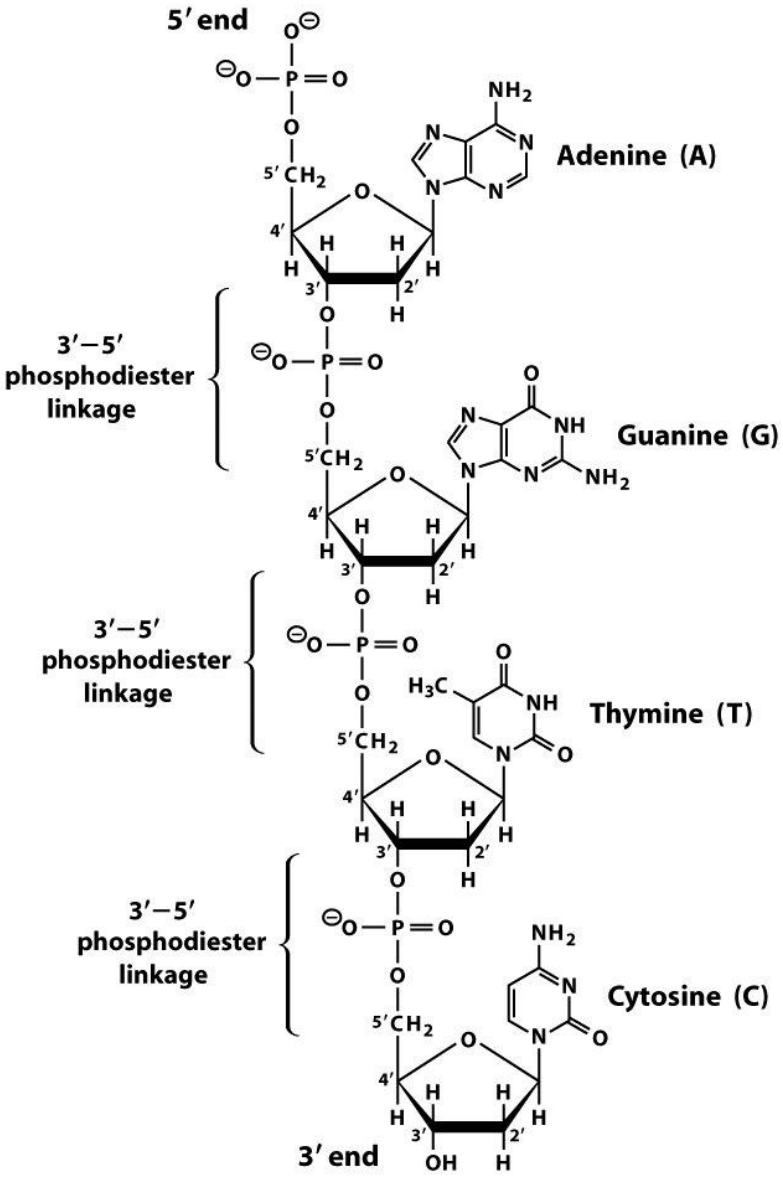
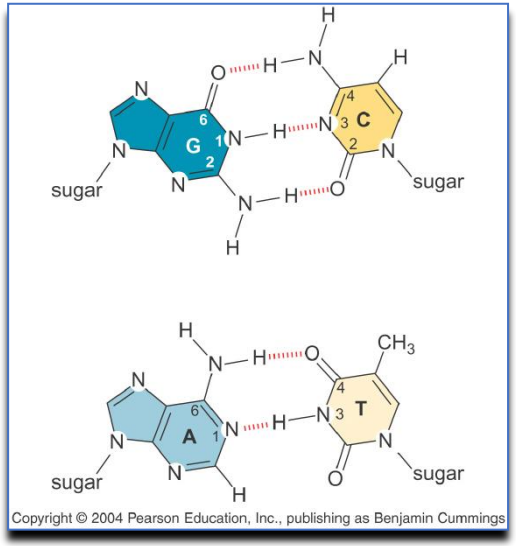


Figure 19-11 Principles of Biochemistry, 4/e
© 2006 Pearson Prentice Hall, Inc.

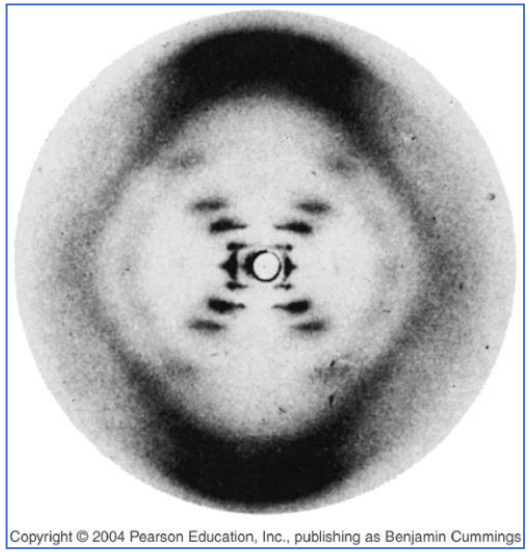
Chargaff's rule:
[A=T], [C=G]
[Purines] =[Pyrimidines]

The double helix

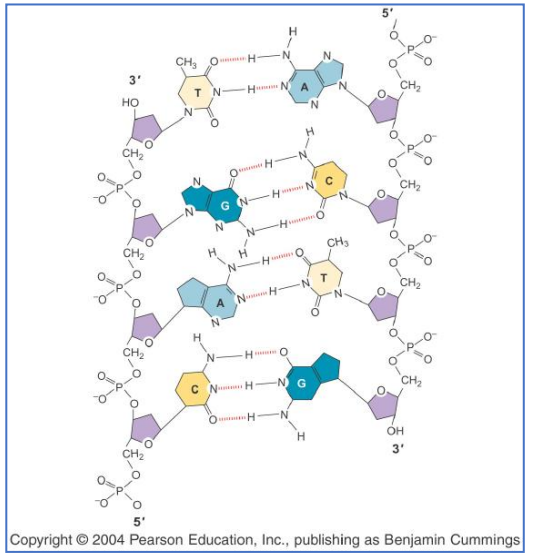


Copyright © 2004 Pearson Education, Inc., publishing as Benjamin Cummings

+


Copyright © 2004 Pearson Education, Inc., publishing as Benjamin Cummings

=


Copyright © 2004 Pearson Education, Inc., publishing as Benjamin Cummings

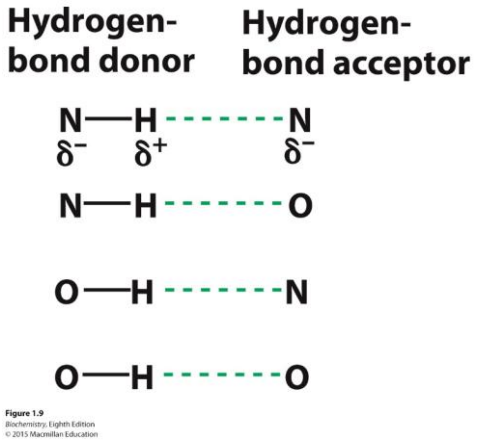


Figure 1.9
Biochemistry, Eighth Edition
© 2015 Macmillan Education

I told them all I knew. If they had heard before about the pairing rules, they concealed it.
*I believe that the double-stranded model of DNA came about as a consequence of our conversation.**

DNA is a double-stranded antiparallel helix

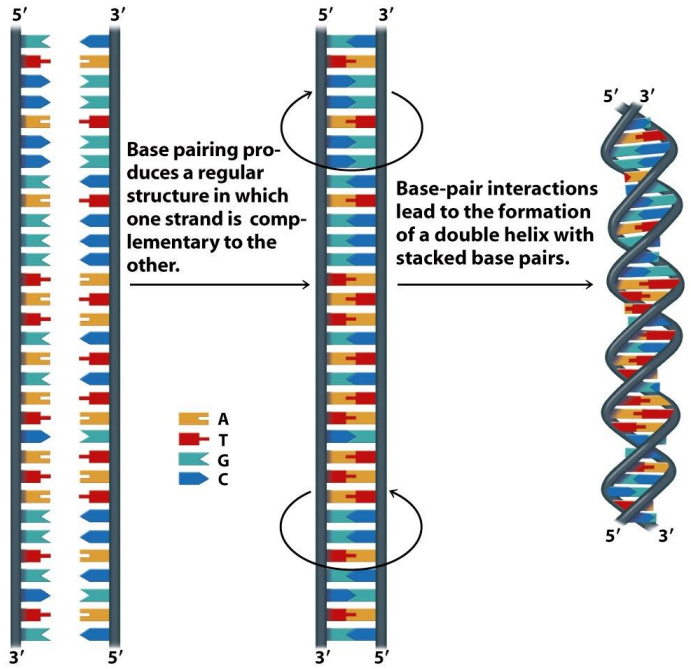
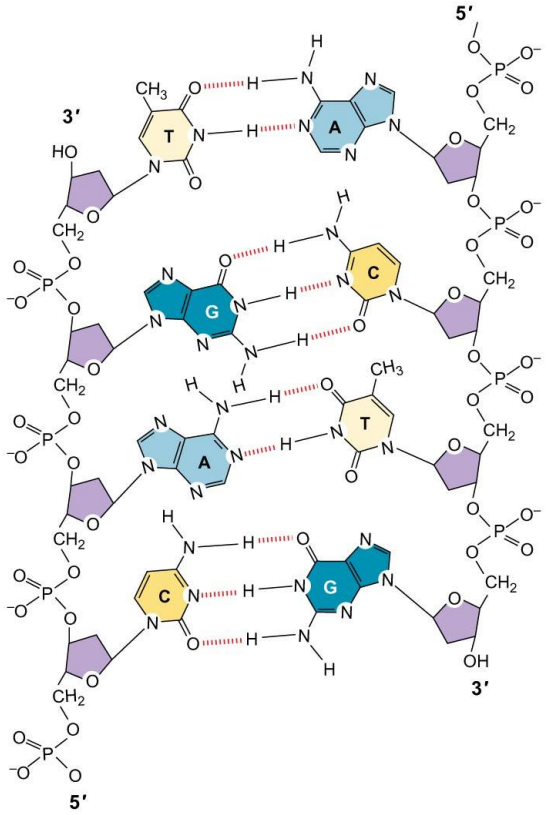


Figure 19-13 Principles of Biochemistry, 4/e
© 2006 Pearson Prentice Hall, Inc.



- DNA is a negatively charged molecule, conferred by the Phosphate groups.
- The Phosphate groups are outward facing and facilitate interactions with water.
- Hydrogen bonds between the nucleotides hold the bases together.
- Base stacking and non covalent interactions such as Van Der Waals forces and electrostatic attractions also contribute to the stability of the helix.

Features and vulnerabilities of DNA and how these are utilised in laboratory molecular biology.

- DNA denaturing and hybridisation
- Susceptibility to nucleases
- Use of restriction enzymes
- DNA in the cell

DNA denaturation and reannealing

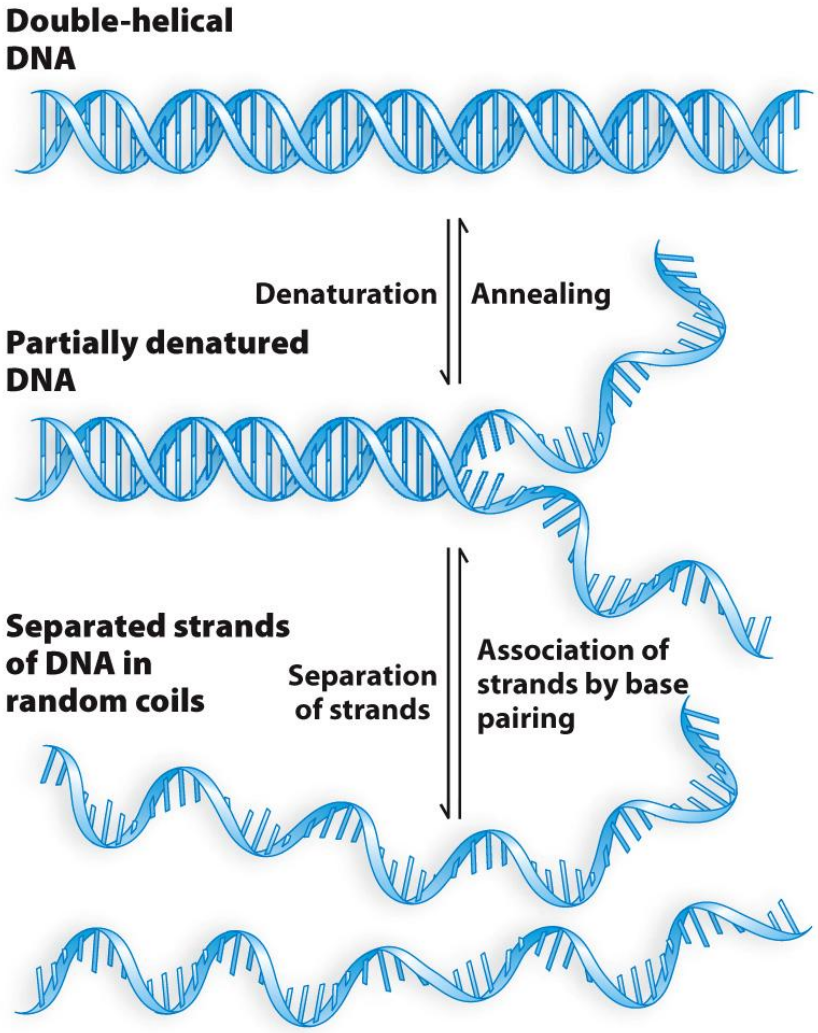


Figure 6-28
Molecular Biology: Principles and Practice, Second Edition
© 2015 Macmillan Education

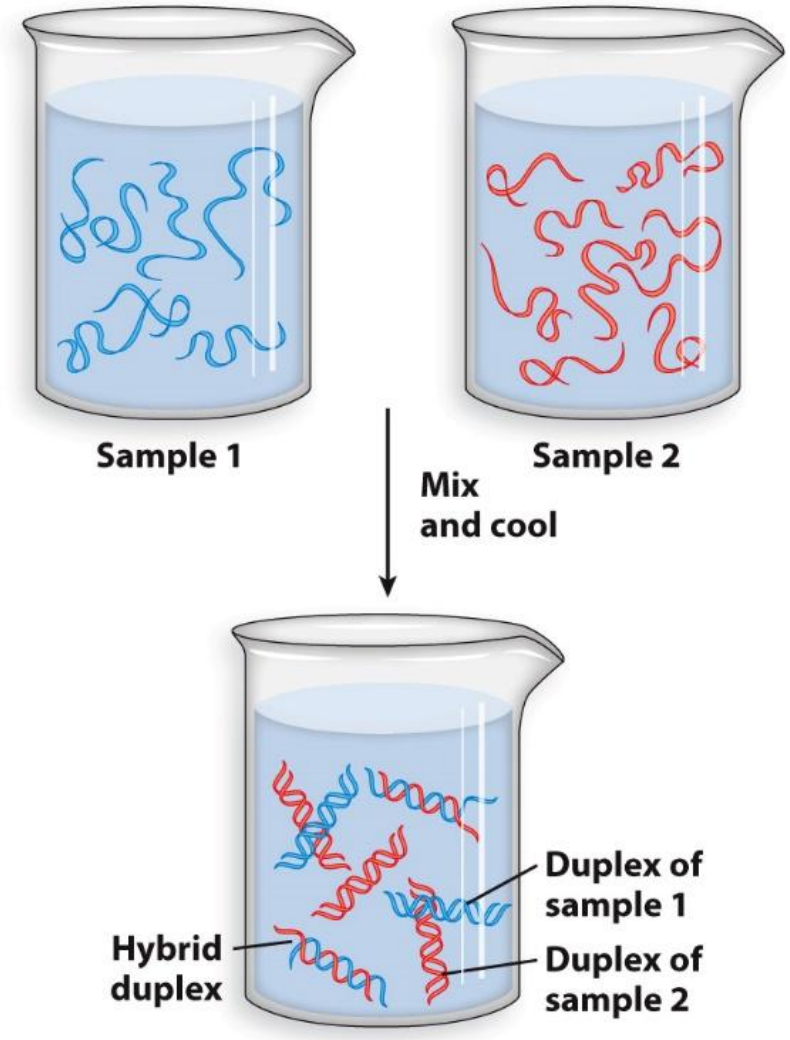


Figure 6-31
Molecular Biology: Principles and Practice, Second Edition
© 2015 Macmillan Education

Nucleases cleave phosphodiester bonds

(A) cleavage of phosphodiester bond leaves product with 5'-phosphate and 3'-hydroxyl ends.

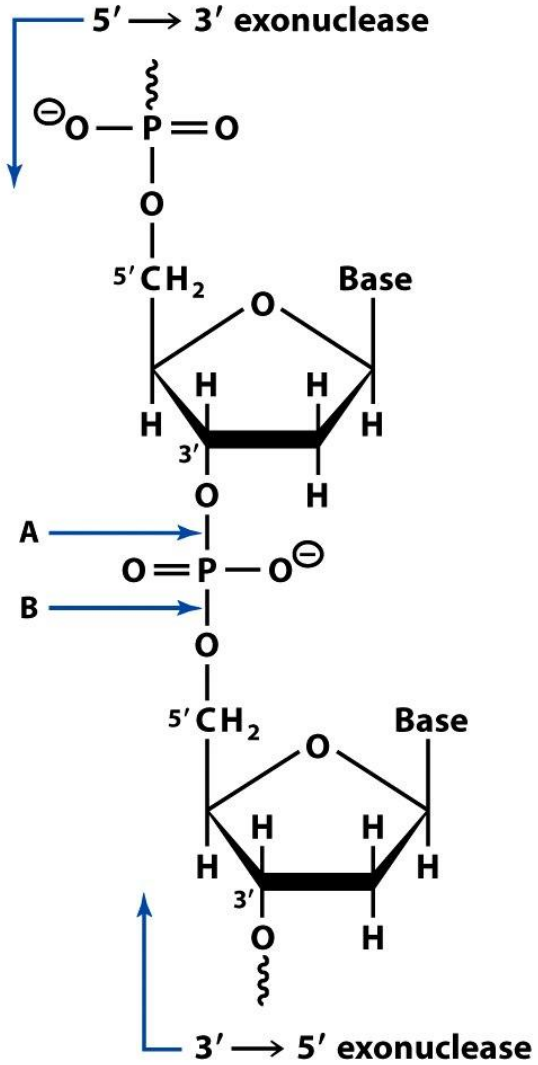


Figure 19-27 Principles of Biochemistry, 4/e
© 2006 Pearson Prentice Hall, Inc.

(B) cleavage of phosphodiester bond leaves product with 3'-phosphate and 5'-hydroxyl ends.

Nucleases cleave phosphodiester bonds

Restriction Enzymes

- Bacterial defence mechanism
- Recognise short sequences of double-stranded DNA as targets for cleavage

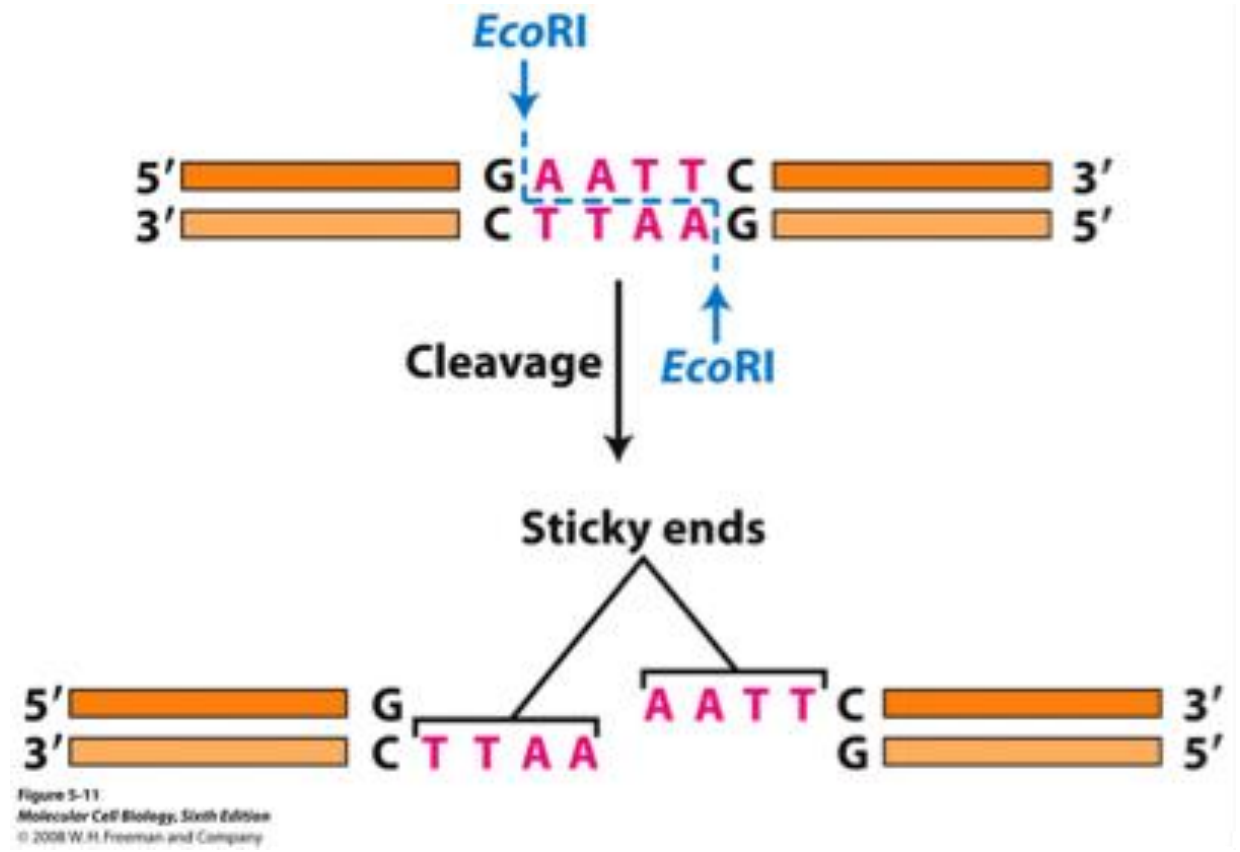
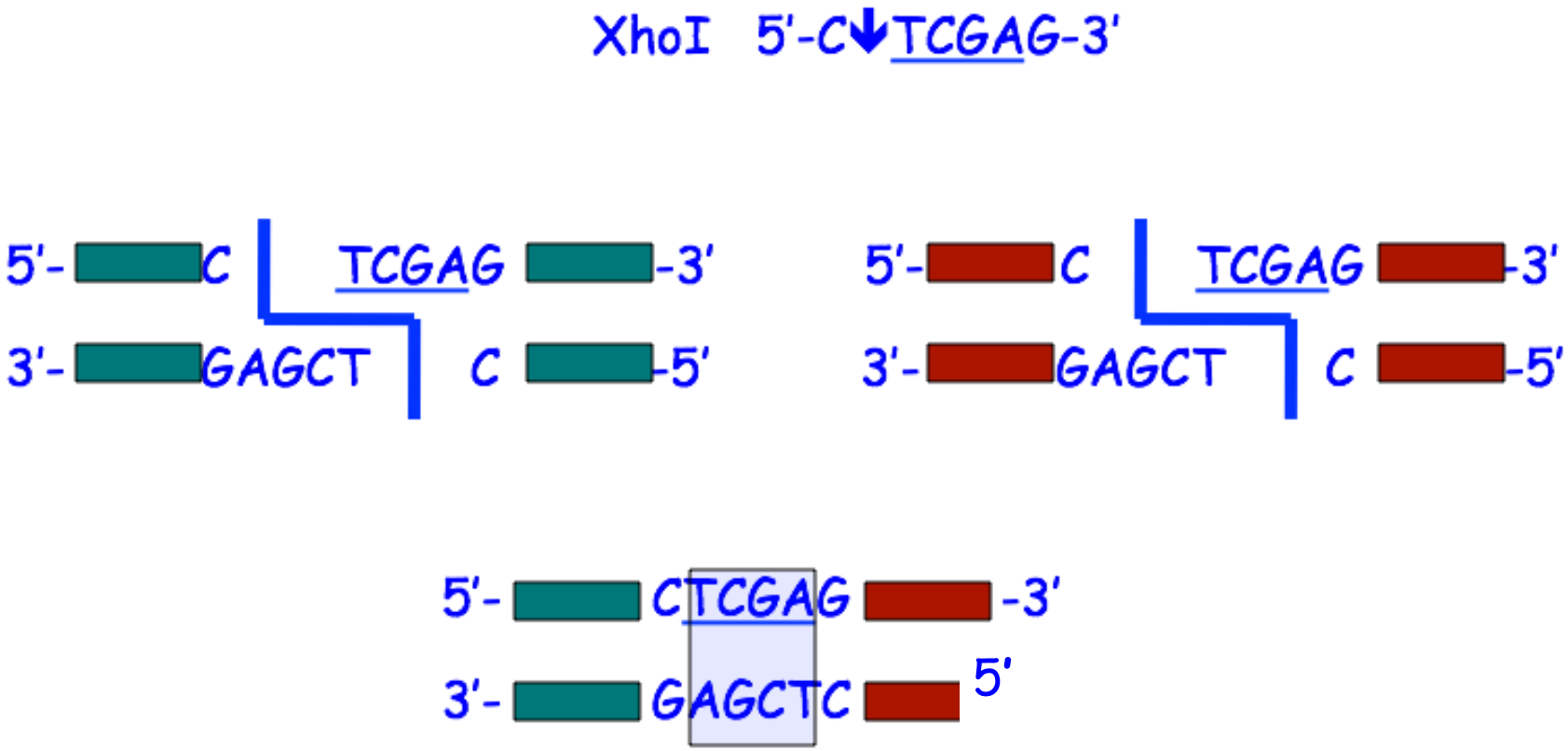


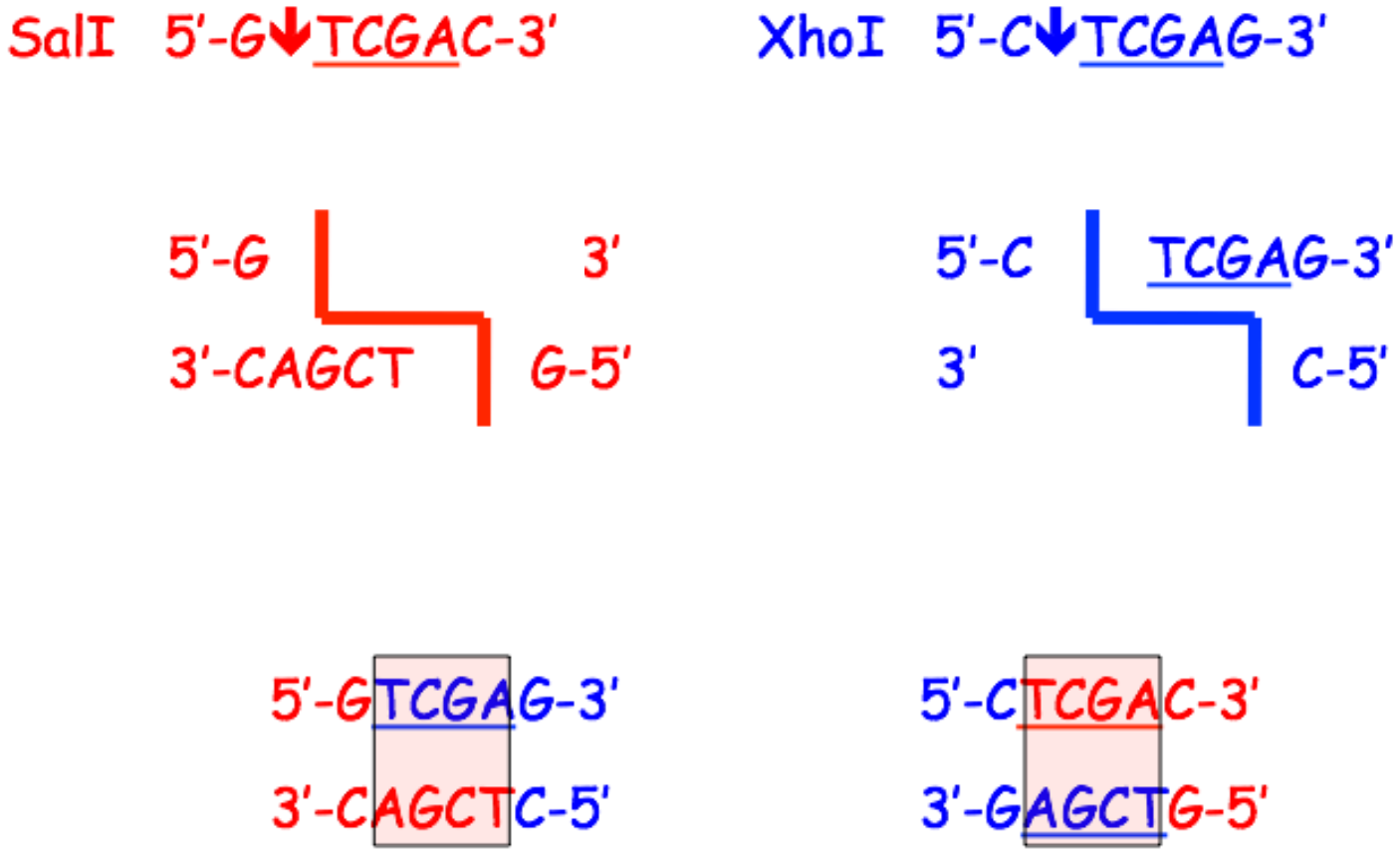
Figure 5-11
Molecular Cell Biology, Sixth Edition
© 2008 W.H. Freeman and Company

Compatible Sticky Ends-joining different DNA strands



© Andrea Townsend-Nicholson 2019

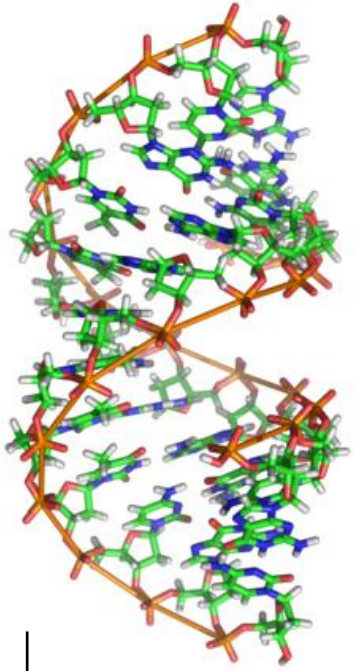
Compatible Sticky Ends-joining different DNA strands using two compatible enzymes

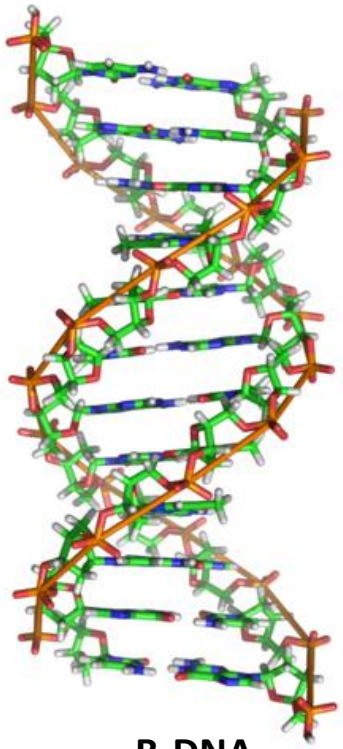


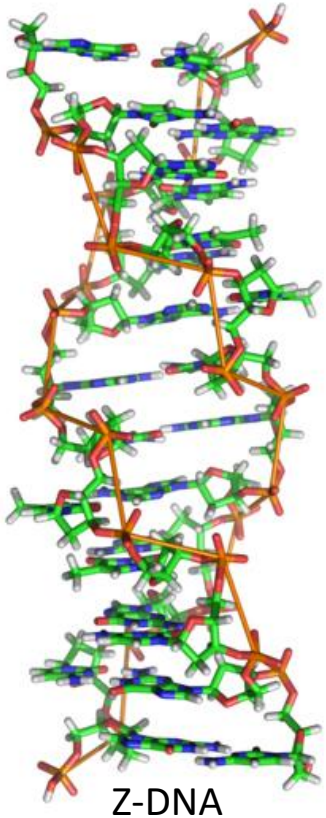
© Andrea Townsend-Nicholson 2019

The DNA Helix: Geometry

**



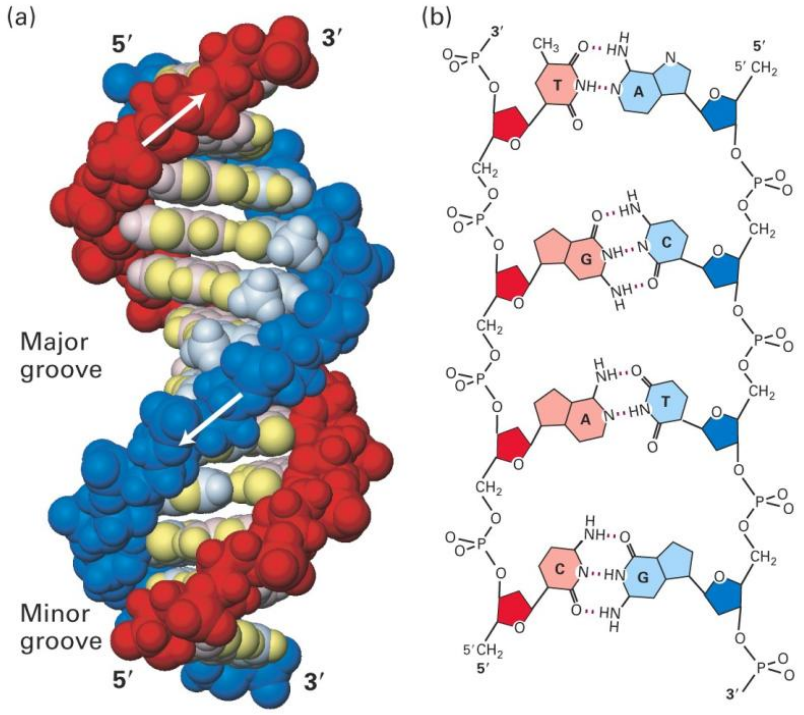




	A-DNA	**B-DNA**	Z-DNA
handedness	right	**right**	left
condition	75% rel. humidity	**92% rel. humidity**	high salt
bp/turn	11	**10**	12
vertical rise (Å/bp)	2.6	**3.4**	3.7
helical diameter in Å (0.1nm)	23	**19**	18

**© Richard Wheeler (Zephyris) http://en.wikipedia.org/wiki/File:A-DNA,_B-DNA_and_Z-DNA.png

B-form DNA



Number of base pairs/turn: **10.4**
Average rise/base pair: **0.34 nm**
Helical diameter: **2.4 nm**
Pitch **3.5 nm**

- The human genome (n) contains 3 x 10^9 base pairs

- If B-DNA has a length of 0.34nm/base pair, how long would the human genome be, if the DNA in your nucleus was laid out base by base?

(answer in the next lecture)!

Take home message

- DNA organization obeys rules:
  - related to charge
  - Related to hydrophobicity
- DNA organization obeys rules related to ability to form bonds
- Interactions of charge, bonds and shape result in the stable double helix
- DNA has susceptibilities-
  - Vulnerability of phosphodiester bonds to endonucleases (restriction enzymes)
  - Vulnerability of hydrogen bonds to heat or alkali-the double helix will separate
- These susceptibilities are utilized in laboratory molecular biology techniques